and 2440 kcal for girls of 13 -15 years and 16-18 years respectively. The difference in energy requirements can be attributed to the rapid increase in weight for boys.

Protein

The protein requirements are computed in the same way as for adults. The protein needs represent 12-14 percent of the total energy requirements. This meets the need for growth, for pubertal changes in both sexes and for developing lean body mass in boys.

Minerals

All mineral needs increase during adolescence. Adolescents at the peak of their growth velocity will require large quantities of nutrients. Adolescents require twice the amount of calcium, iron, zinc, magnesium and sodium during the years of the growth spurt compared with that during other years. Bone growth requires calcium. During adolescence 800 mg of calcium per day is needed. About 150 mg of calcium should be retained for increasing bone mass. Adolescents who have less bone mineral density are susceptible for osteoporosis later in their life.

The requirements for iron have been arrived at 32 mg/day and 27 mg/day for boys and girls 13 – 15 years of age, and 28 mg / day and 26 mg / day for boys and girls 16-18 years of age respectively. Iron requirements are generally high during adolescence due to the following reasons such as growth spurt, expansion of blood volume, increase in haemoglobin concentration, additional iron to compensate menstrual losses in girls and adding the basal loss of iron. Zinc supplements can also increase pubertal growth in adolescents.

Vitamins

The requirement for B vitamins namely thiamine, riboflavin and niacin increases in direct proportion with increase in calorie intake. Folic acid and vitamin B12 requirements also increase when there is rapid tissue synthesis as they participate in synthesis of DNA and RNA.

Transamination to synthesize non-essential amino acids requires more vitamin B6. The structural and functional integrity of newly formed cells depends on the availability of vitamins A, C and E

3.5.4 NUTRITIONAL PROBLEMS

Obesity

Obesity is a state in which there is a generalized accumulation of excess fat in adipose tissue in the body leading to more than 20% of desirable weight. Obesity has several adverse health effects and can even lead to premature death. Obesity leads to high blood cholesterol, high blood pressure, heart disease, diabetes, gallbladder stone and certain types of cancer.

Factors responsible for adolescent obesity

- Inactive life style
- Genetics and family history
- Junk food and over eating
- Psychological causes
- Environmental causes
- Diseases
- Drugs
- Tobacco and alcohol

The prevalence of adolescent obesity can be reduced by initiating programmes of health and nutrition education for school children, encouraging physical activity and healthy food habits.

Some of the factors contributing to adolescent obesity and solutions to prevent it are presented in Table 3.4

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